

October 27, 2025

Master Degree in Bionics Engineering

Course of Principles of Bionics and Biorobotics Engineering

Lesson title:

Swimming

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THE BIORBOTICS
INSTITUTE



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Swimming

Physics of swimming

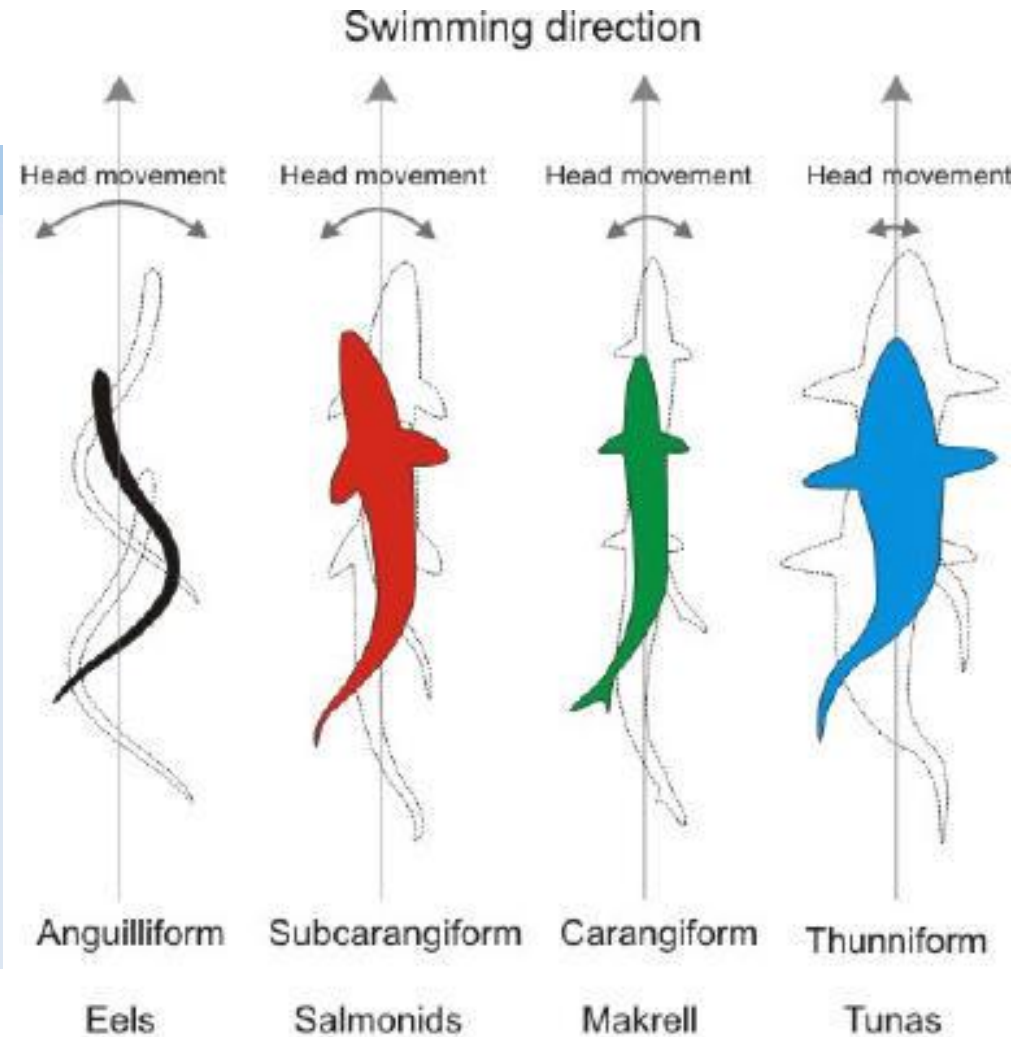
Properties of water as medium in which to move:

- density - 830x greater than air
- viscosity -70x greater than air
- lift
- drag - 830x greater than in air
 - increases with speed of object or current
 - due to separation of flow from object into turbulent flow
- boundary layer
 - laminar or turbulent



body/caudal fin propulsion

1. anguilliform: eels, sea snakes
undulate entire body, largest head displacement (same as snake 'serpentine movement')
2. subcarangiform: cod, salmonids: undulation involves more body than just peduncle and tail
3. carangiform: oscillating only tail fin and peduncle
4. Tunniform: tunas etc.



Sfakiotakis, M., Lane, D. M., & Davies, J. B. C. (1999). Review of fish swimming modes for aquatic locomotion. IEEE Journal of Oceanic Engineering, 24(2), 237-252.

Anguilliform locomotion



Thunniform locomotion



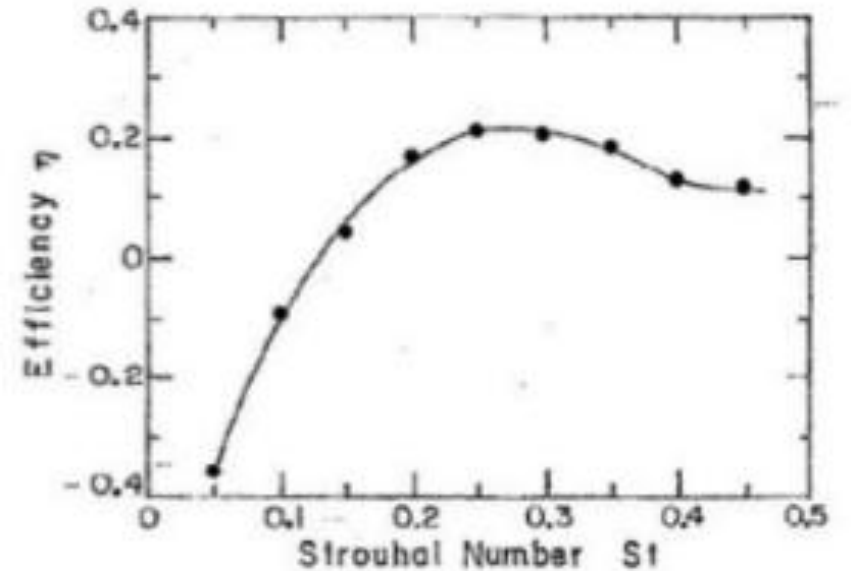
Swimming

The **Strouhal Number** is a dimensionless number describing oscillating flow mechanisms. The Strouhal number is an integral part of the fundamentals of fluid mechanics.

$$St = \frac{fL}{V}$$

The Strouhal number is often given as St dimensionless Strouhal number, f frequency of vortex shedding, L characteristic length (for example hydraulic diameter) V velocity of the fluid. A Strouhal number is the principle parameter in an oscillating foil.

SWIMMING MODE	SPECIES	PROPULSIVE (FROUDE) EFFICIENCY
Anguilliform	Eel	0.43 – 0.54
Carangiform	Rainbow Trout	0.74
Thunniform	False Killer Whale	0.85 – 0.91



The **Froude number** is based on the speed–length ratio:

$$Fr = \frac{u_0}{\sqrt{g_0 l_0}}$$

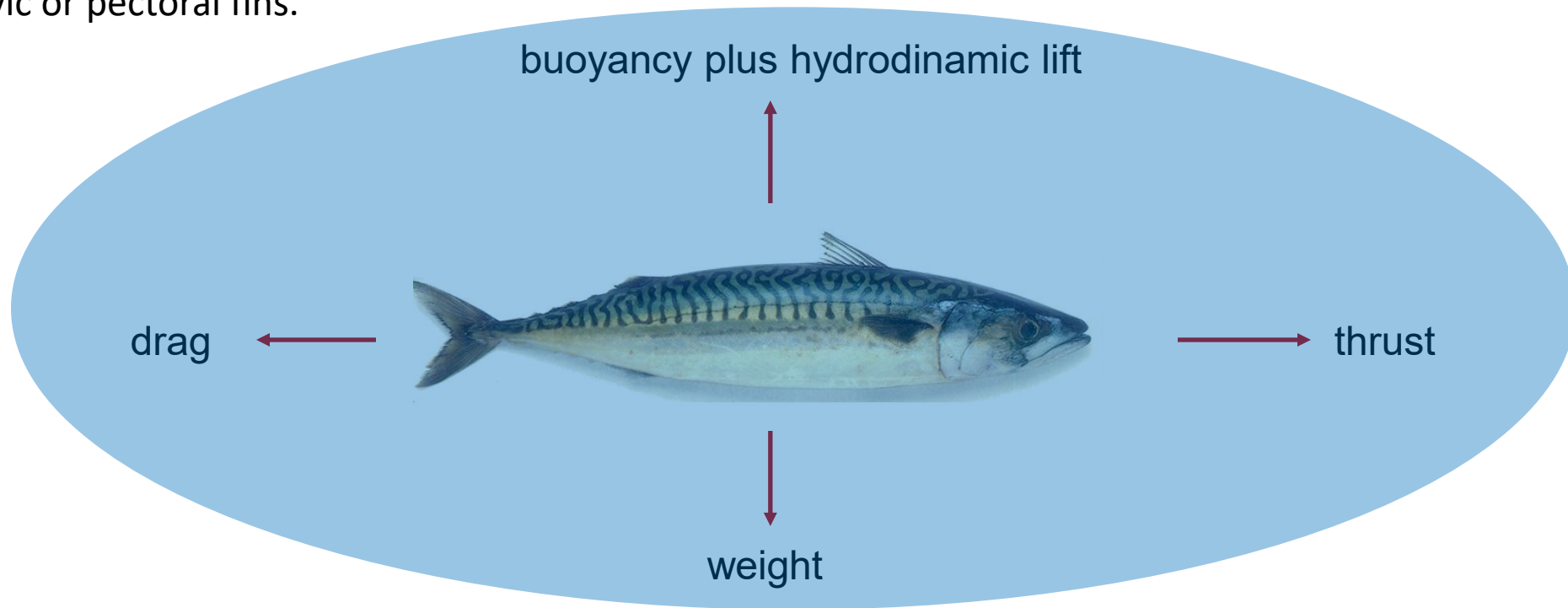
where u_0 is a characteristic flow velocity, g_0 is in general a characteristic external field, and l_0 is a characteristic length.

It is a very significant figure used to determine the resistance of a partially submerged object moving through water.

Swimming

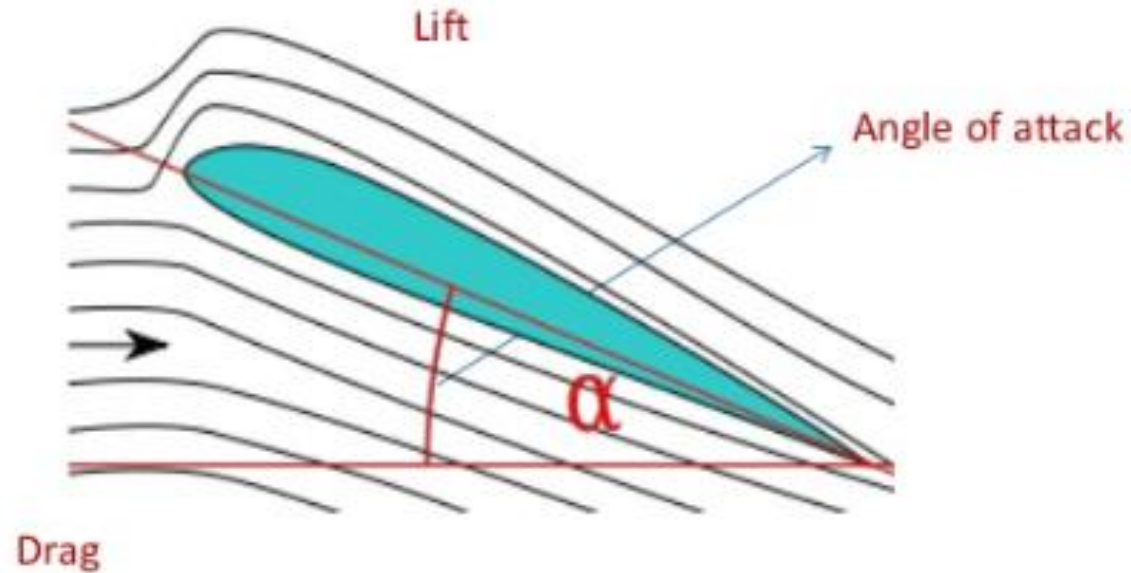
Forces acting on a swimming aquatic animal

- In the **horizontal direction**, propulsive thrust is the force that drives the aquatic vertebrate forward and is a function of the work output from the fish or marine mammal.
- **Drag** opposes propulsive thrust contributing to swimming inefficiencies.
- In the **vertical direction**, the weight of the fish is acting downward counteracted by buoyancy and lift forces acting upward.
- The **lift force**, which too may act in the downward direction, is typically related to the lift generated during swimming due to the pelvic or pectoral fins.



Swimming

Angle Of Attack is a term used in fluid dynamics which describes the angle between the lifting body's reference line and the oncoming flow.



Along the Direction of Flow – Drag
Perpendicular to the Direction of Flow - Lift

Swimming

Reynolds number (Re): ratio of inertial forces to viscous forces within a fluid which is subjected to relative internal movement due to different fluid velocities, which is known as a boundary layer in the case of a bounding surface such as the interior of a pipe.

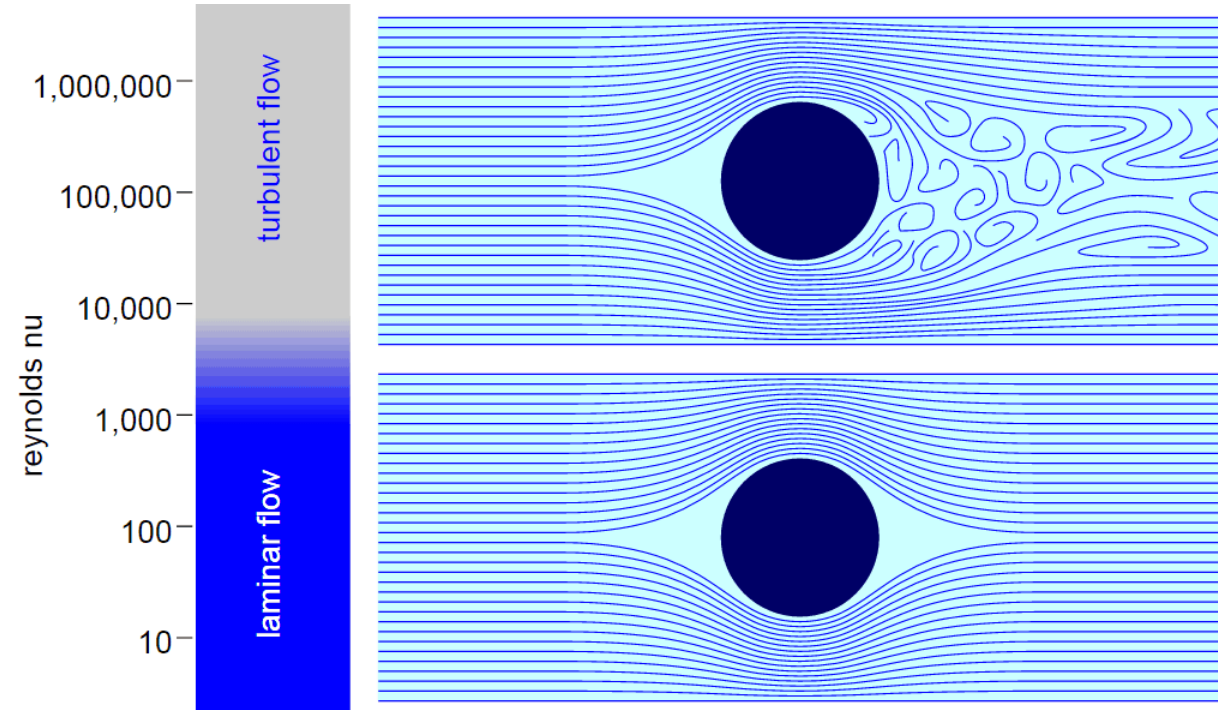
$$Re = \rho u L / \mu$$

ρ is the density of the fluid (SI units: kg/m³)

u is the velocity of the fluid with respect to the object (m/s)

L is a characteristic linear dimension (m)

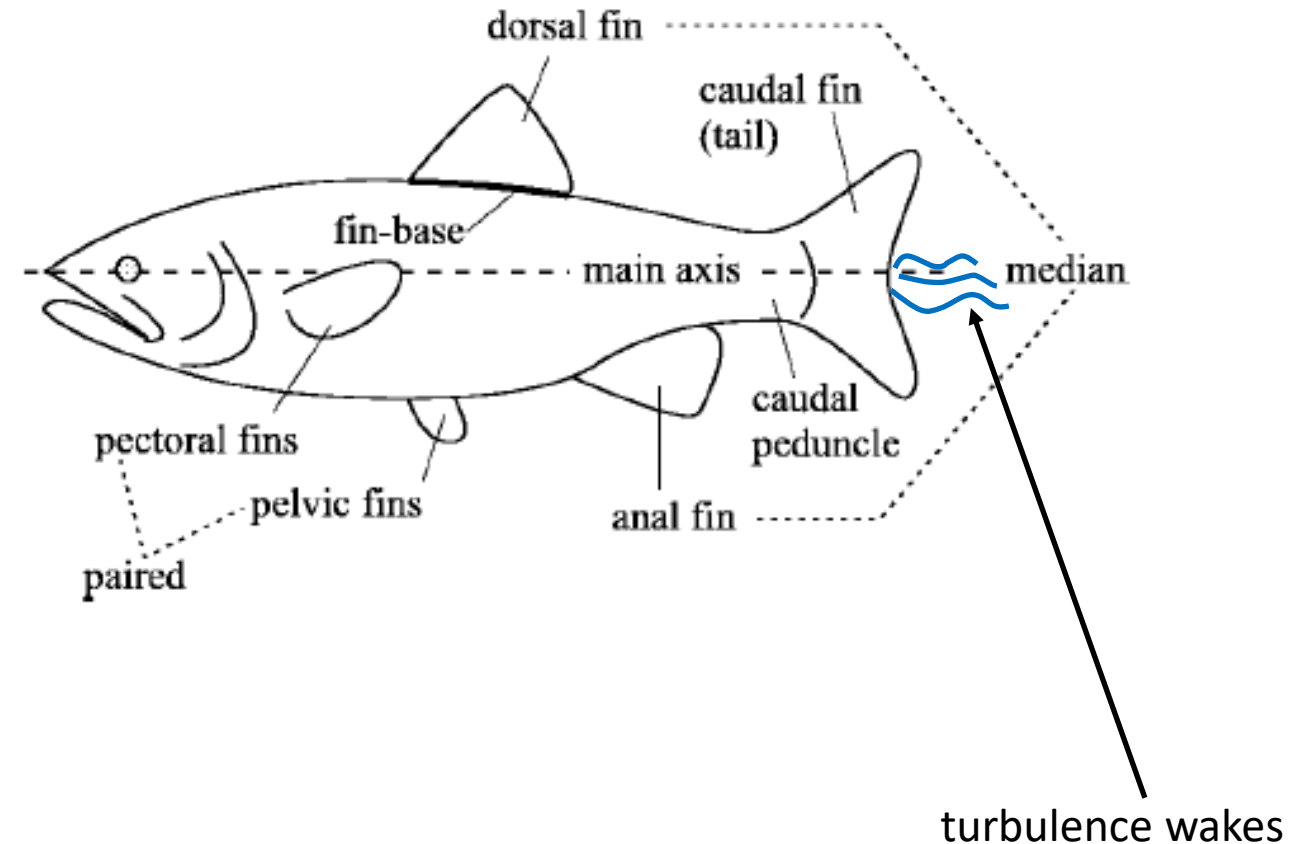
μ is the dynamic viscosity of the fluid (Pa·s or N·s/m² or kg/m·s)



Swimming

Increasing swimming efficiency:

- avoid separation of boundary layer from surface
- maximize laminar flow in boundary layer
- minimize turbulent flow in wake



<https://www.youtube.com/watch?v=XIBVQoRGilw>

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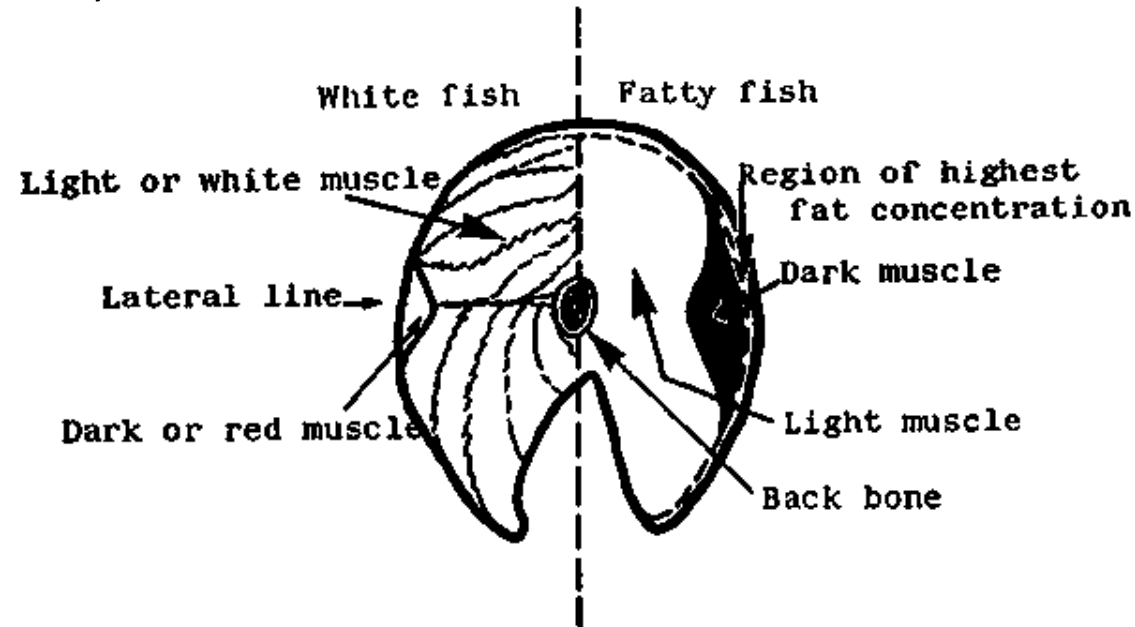
Swimming

active swimming is accomplished using **red muscle** along sides of fish (sustained for hours or days)

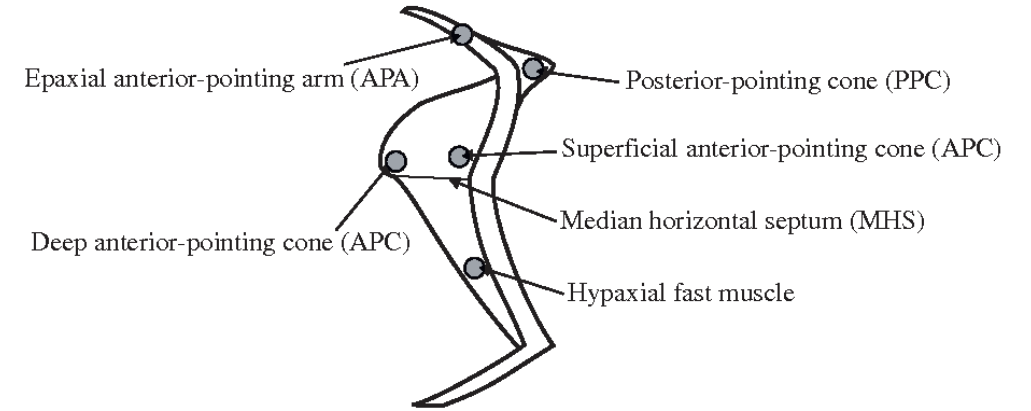
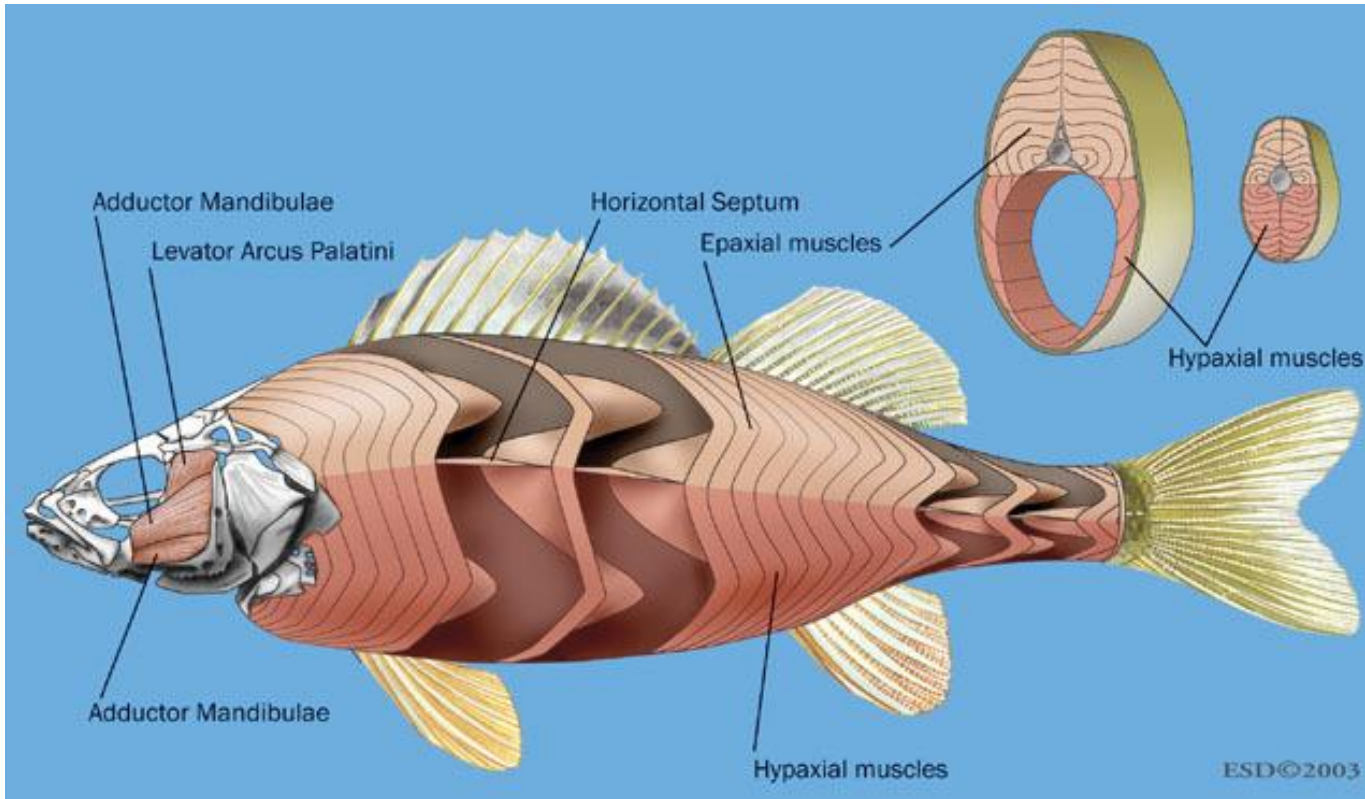
- high myoglobin and mitochondrial enzymes

burst swimming is accomplished using **white muscle** (only for up to 30 secs)

- great contractile speeds, low endurance

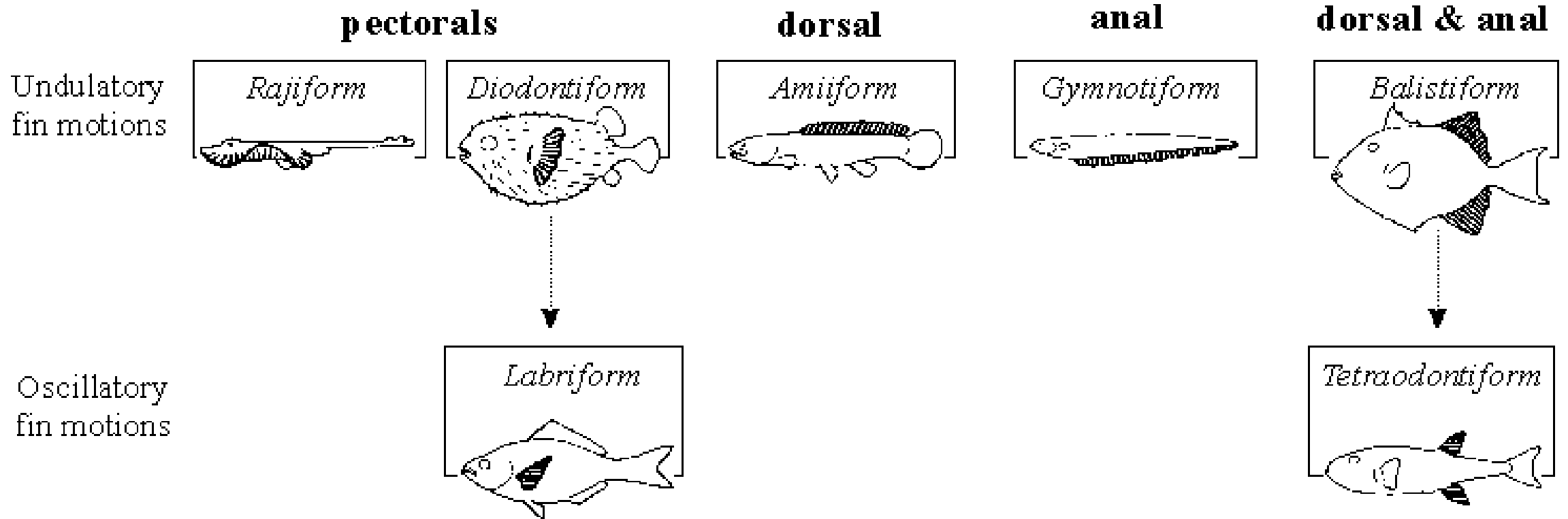


Fish do not have muscles arranged in bundles (e.g. like terrestrial vertebrates). Fish have their muscles laid in sheets called **myotomes**. Each myotome is separated from the one next to it by connective tissue. Myotomes are folded into a 3-D shape.



At the lateral midline they are divided by a horizontal septum into dorsal (epaxial) and ventral (hypaxial) regions.

MPF - median/paired fin propulsion



Lateral line

The fish lateral line is a system of sense organs used to detect movement, vibration, and pressure gradients in the surrounding water.

The physical properties of the lateral line and its significance in flow sensing is an active research topic in biorobotics and bionics.

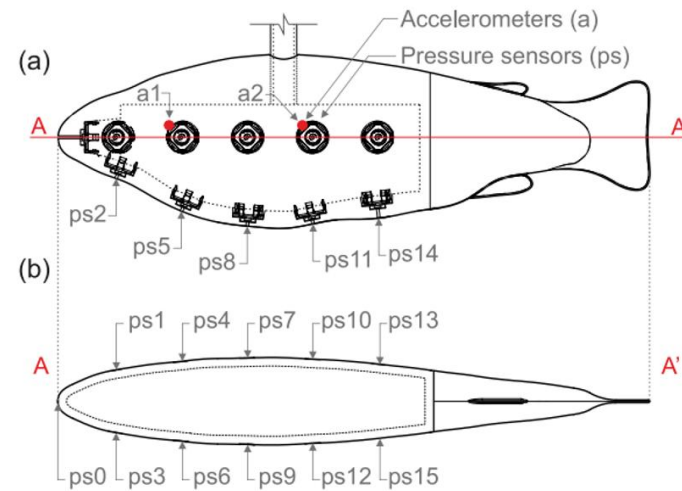
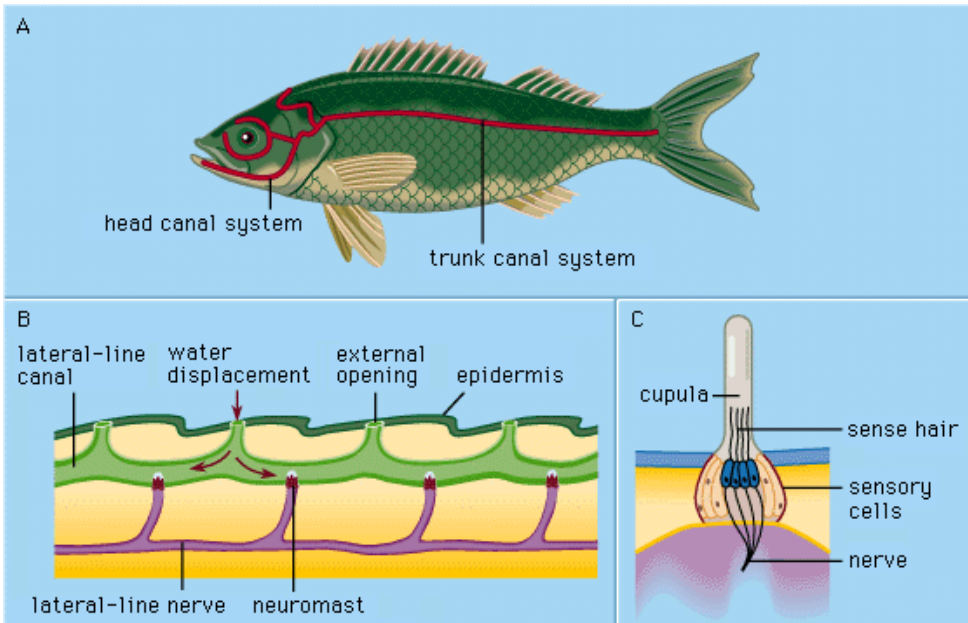


FIG. 1. Schematic of the fish-shaped lateral line probe. (a) Side view showing bottom and 5 of the 10 lateral pressure sensors and inline accelerometers. (b) Top view showing all 16 pressure sensor locations. In this work, only the nose sensor ps0 in conjunction with the 10 lateral sensors was used.

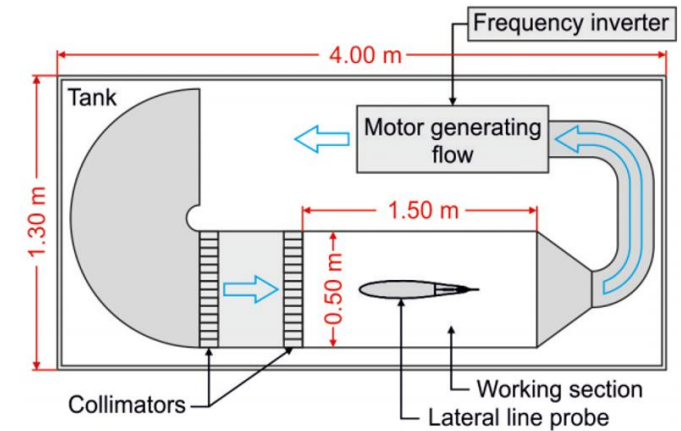


FIG. 2. Experimental setup in the closed flow tunnel (experiment 1), view from above. The LLP was placed in the center of the flow tunnel working section and subjected to a uniform freestream velocity field (flow is from left to right). Homogeneous turbulence in the upstream flow is produced by passing the water through two collimators placed directly upstream of the working section.

Tuhtan, J. A., Fuentes-Pérez, J. F., Strokina, N., Toming, G., Musall, M., Noack, M., ... & Kruusmaa, M. (2016). Design and application of a fish-shaped lateral line probe for flow measurement. *Review of Scientific Instruments*, 87(4), 045110.